Recurrent Attention and Semantic Gate for Remote Sensing Image Captioning

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***Abstract*— The remote sensing image captioning has attracted wide spread attention in remote sensing field due to its application potentiality. However, most existing approaches model limited interactions between image content and sentence and fail to exploit special characteristics of the remote sensing images. We introduce a novel recurrent attention and semantic gate (RASG) framework to facilitate the remote sensing image captioning in this article, which integrates competitive visual features and a recurrent attention mechanism to generate a better context vector for the images every time as well as enhances the representations of the current word state. Specifically, we first project each image into competitive visual features by taking the advantage of both static visual features and multiscale features. Then, a novel recurrent attention mechanism is developed to extract the high-level attentive maps from encoded features and nonvisual features, which can help the decoder recognize and focus on the effective information for understanding the complex content of the remote sensing images. Finally, the hidden states from the long short-term memory (LSTM) and other semantic references are incorporated into a semantic gate, which contributes to more comprehensive and precise semantic understanding. Comprehensive experiments on three widely used datasets, Sydney-Captions, UCM-Captions, and Remote Sensing Image Captioning Dataset, have demonstrated the superiority of the proposed RASG over a series of attentive models based on image captioning methods.**

***Index Terms*— Attention mechanism, encoder-decoder, remote sensing image captioning, semantic understanding.**

NOMENCLATURE

I	The input remote sensing image.
K	Number of parallel dilated convolutional branches.
D	Dimension of each feature.
C	Dimension of each feature.
$H \times W$	The counts of image location features.
V_{CNN}	Feature map from "conv5_3" layer of the VGG16.
v_k	The kth image location features in V_{CNN}.
$\bar{v}$	The global feature computed from V_{CNN}.
V_{ESP}	Feature map from the ESP module.
v'_k	The kth image location features in V_{ESP}.
G	The global feature computed from V_{ESP}.
V_{enc}	Feature map from the Feature Extractor.
h_t	Word state at t time.
w_t	Word embedding at t time.
y_t	The generated word at t time.
x_t^1	Input for the Attention LSTM at t time.
h_t^1	Hidden state of the Attention LSTM at t time.
g_t^1	The gate vector of the Attention LSTM at t time.
m_t^1	The memory vector of the Attention LSTM at t time.
s_t^1	The "visual sentinel" vector of the Attention LSTM at t time.
h_{t-1}^1	Hidden state of the Attention LSTM at $t-1$ time.
α_t	The attention weights at t time.
$\alpha_{i,t}$	The ith attention weight at t time.
u_t	The attention scores distributed in V_{enc} at t time.
att_t	Attentive result from the Recurrent Attention at t time.
att_{t-1}	Attentive result from the Recurrent Attention at $t-1$ time.
H_t	New current hidden state at t time.
x_t^2	Input for the Attention LSTM at t time.
h_t^2	Hidden state of the Language LSTM at t time.
g_t^2	The gate vector of the Language LSTM at t time.
m_t^2	The memory vector of the Language LSTM at t time.
s_t^2	The "visual sentinel" vector of the Language LSTM at t time.
h_{t-1}^2	Hidden state of the Language LSTM at $t-1$ time.
c_s	The supplementary semantic representation at t time.
z_t	An intermediate semantic vector at t time.
$\tilde{h}_t^2$	The word state for the Semantic Gate at t time.
p_t	The probability over the vocabulary at t time.
Y	The ground-truth caption.
y_i	The ith word in the ground-truth caption.
S	The generated caption.
s_i	The ith word in the generated caption.

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I. Introduction

DIFFERENT from image classification [1], [2], object detection [3], and semantic segmentation [4], [5], the remote sensing image captioning aims at translating the remote sensing images as natural language sentences, which has attracted more and more attention in recent years. It requires to understand the visual content [6] as comprehensive as possible and model the semantic relations between the objects with the natural language [7], [8]. Fig. 1 shows a captioning example of an image from Remote Sensing Image Captioning Dataset (RSICD) [9] dataset. As depicted in Fig. 1, the visual content clearly matches "two tennis courts," "baseball field," "parking lot," "open space," "green trees," "roads," and "buildings," and the semantic relation matches "in," "between," "beside" and "are surrounded by."

The generated sentences can provide help for people in many fields, e.g., disaster monitoring, resources survey, as well as military intelligence. Similar to the image captioning [10], [11] for the natural image taken by a digital camera, video captioning [12]–[14] and visual question answering (VQA) [15]–[17], the remote sensing image captioning is known as a challenging task, since its unique features and complicated interactions between the visual content and the natural language obstruct the development of the visual content understanding. It is noted that the image captioning for the natural image is described as the natural image captioning in the following.

The existed natural image captioning methods include mainly three general paradigms: 1) the template-based paradigm [18], [19] that requires knowing a set of words to describe the salient objects of an image and embeds them into a designed sentence template; 2) the retrieval-based paradigm [20], [21] which computes the similarity between the images and the captions, and then matches the known well-formed sentences according to higher similarity; and 3) the encoder–decoder paradigm [11] that extracts the contextual representation of the image and translates the contextual feature as the natural language.

However, a template-based approach suffers from the scarcity of diversity in generated sentences, and a retrieval-based approach may be generated inappropriate sentences due to the limitation of precollected sentence set. Inspired by the neural translation machine [22], [23], the encoder–decoder paradigm is applied to natural image captioning and mostly belongs to the "convolutional neural network-recurrent neural network (CNN-RNN)" architecture, which not only learns simultaneously image representation encoder and sentence decoder, but also outperforms the mentioned traditional approaches. Moreover, the researchers found that the visual attention mechanism is helpful for a machine to understand image content. Specifically, the attention mechanism was first employed in natural image captioning by Xu *et al.* [24], which focused on the most salient visual regions and weakened the other parts. Later, Lu *et al.* [25] came up with an adaptive attention mechanism, which made the visual and nonvisual information more compatible with a "visual sentinel" than individual concepts



(1) Two tennis courts, a baseball field and a parking lot are in the open space between crowded buildings
(2) A irregular court and a parking lot beside.
(3) A baseball field and two tennis courts are between two piece of buildings
(4) A baseball field and two tennis courts are between two roads and several buildings
(5) Two tennis courts and a baseball field are surrounded by some green trees and many buildings

Fig. 1. Example of remote sensing image content description. Five ground truth descriptions are exhibited in the right.

derived from the images. Also using attention mechanism, the key benefit of the top-down and bottom-up attention mechanism [26] was that the bottom-up features were based on object detection regional outputs, and the designed language modeler was conducive to describe precisely the image. Recently, the proposed attention module [27], [28] facilitated both visual feature enhancement and multimodal feature interaction for the image captioning by further measuring the relevance between the attention result and the query. The attention-based methods are beloved by all researchers on the vision-to-language (V2L) tasks. Unfortunately, directly integrating the natural image captioning models in the remote sensing image captioning work may be too-coarse to leverage the image information for semantic understanding. Most intuitively, there are many differences between the natural image and the remote sensing image expression. The special characteristics of the remote sensing images require the image captioning model to be capable to analyze dynamically different types of scenic, such as the numerous target categories, changeable scales, etc. Therefore, the problem of how to exploit a suitable structure is of great significance for remote sensing image captioning remains pretty much open.

Many researchers have proposed various neural network models to achieve the content understanding of the remote sensing image by leveraging image semantic information. An earlier approach proposed by Zhang *et al.* [29] built the semantic feature embedded to long short-term memory (LSTM) through a classifier for the input images. However, it lacks the spatial features of objects and is very important for the image caption. A refined solution was introduced by Zhang *et al.* [30] to apply the image attributes to correlate a specific image region using an attribute attention mechanism. The method is able to refine the visual details by combing the attribute-attention correspondences and achieves a lot of gratifying results. To make the semantic attributes more precisely, the label-attention mechanism (LAM) [31] was constructed to compute the attention weights with the prosperous effect of specific labels, which enhanced the effectiveness of the contextual vectors. But it is not an end-to-end model. Li *et al.* [32] noticed that many previous attention mechanism mainly aimed at the remote sensing images, so they proposed multilevel attention (MLA) mechanism decomposed the generation process into a visual attention step, a history semantic attention step, and a contextual layout constructing step with attention. Additionally, Zhao *et al.* [33]

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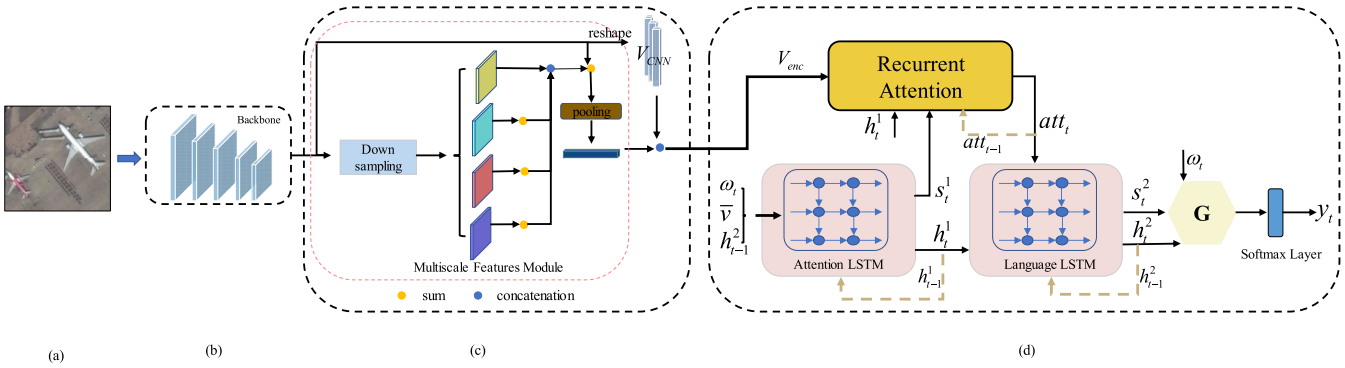


Fig. 2. Overall flowchart of our proposed RASG method. The original CNN features of remote sensing images are first extracted and feed them into the ESP module for multiscale features. The feature extractor combines the original CNN features with the global feature of multiscale features to make up the competitive feature V_{enc}. In decoder module, the competitive features are modeled by the recurrent attention, the Attention LSTM and Language LSTM are used to preliminarily transform visual information to semantic information, and the semantic gate finally generate the deliberate caption. (a) Input image. (b) Feature map. (c) Feature extractor. (d) Decoder module.

designed a structured attention mechanism to exploit the semantic contents in the remote sensing images for caption generation and image segmentation which provided structured geometry and appearance. Instead of only employing visual inputs, Lu *et al.* [34] proposed a sound active attention (SAA) framework which effectively utilizes sound and visual cues. To obtain more direct and advanced semantic information guidance, Sumbul *et al.* [35] summarized the ground-truth captions into a single caption which contained the semantic content of the image. Wang *et al.* [36] proposed a word-sentence framework to summarize the visual content by a word extractor, the word for the decoder module is more suitable than visual features which may have insufficient semantic information to specify. Recently, the idea of reinforcement learning [37] was found to be beneficial for the image captioning as well. So Shen *et al.* [38] proposed a Variational Autoencoder and Reinforcement Learning based Two-stage Multitask Learning Model (VRTMM) to utilize both high and low features, which can generate better description quality. Other works like [39] proposed a truncation cross entropy for relieving the overfitting problem caused by the lack of remote sensing images. Experimental results demonstrated that these neural architectures have achieved remarkable performance improvements in the remote sensing image captioning. Nevertheless, the previous models still have the following limitations.

1) Most researchers started from extracting the image features by CNNs, and then shifted gazes to regional image features with various attention mechanisms. Finally, a language module is used to generate the caption. Apparently, the existing models obtained image features without expressing the multiscale information of the remote sensing images, which is very important for the remote sensing image captioning.
2) A better context vector is helpful for exploring deeper semantic information of the image. In particular, the adaptive attention can well leverage the visual and nonvisual information, but it deviates from the fact that there is a contextual connection among the sentence generation. So, striving for contextual relevance remains a tough challenge.
3) Although the decoder can provide the heuristically word state h_t for the text generation in most existing methods, the predicted word is highly dependent on the word state h_t. When predicting one error word at a time, the decoder can easily cause the accumulated error in final generated sentences. Therefore, it is necessary to learn more semantic reference information at each timestep to result in an informative word state, which is essential for narrowing the gap between the images and text.

Considering above factors, we design the recurrent attention and semantic gate (RASG) net, a novel architecture that learns robustly a superior image captioning model from the RSICDs in this article. The proposed framework is displayed in Fig. 2. At first, for addressing the limitations of the encoder module in existing methods, the efficient spatial pyramid (ESP) [40] is used to extract the multiscale features from the spatial features obtained by a pretrained CNN, which are fed to an adaptive-avg layer for a global representation. Meanwhile, the original CNN features, as well as the global representation associated with the input image, are constructed to fusion features through concatenation. However, the noneffective attentive maps will deteriorate the performance of image captioning models. Therefore, a recurrent attention mechanism is presented in this article, which can capture the high-level attentive maps and divide the attention results into visual and nonvisual information. Using the previous attentive result with plentiful contextual relevance, salient visual features are preserved adaptively while nonvisual features (i.e., the "visual sentinel" vectors from the Attention LSTM) are weakened. Thus, the Language LSTM can be enhanced with the help of the recurrent attention mechanism. Moreover, we designed a semantic gate to embed deeper semantics for sequential word generation by adding other semantic reference information. The semantic vectors obtained by the semantic gate contain effective and efficient textual information. Finally, extensive experiments on three benchmark datasets (Sydney-Captions,

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UCM-Captions and RSICD) illustrate the superiority of our proposal compared to other state-of-the-art approaches.

Overall, the features of the proposed RASG model can be summarized in the following four aspects.

1) Our framework combines an improved attention mechanism with competitive visual feature construction to effectively encode the input image into a context-aware feature representation, while the strengthened word states for generating the output sentence are modeled through a learnable semantic gating mechanism.
2) In our visual encoder, the original CNN features are employed in a multilevel way to learn the multiscale features of the remote sensing images. Integrated with the CNN features, the entire system benefits from not only the static visual representation, but also multiscale features that are modeled via different dilated convolutions.
3) In contrast to many designed attention models passively avoiding textual elements that represent the potential relations among objects, the proposed recurrent attention directly learns distributions from encoded visual features and latent nonvisual features. Further, the recurrent attention theoretically takes contextual relevance into account.
4) To avoid wrong prediction in generated captions, we devise a learnable semantic gate that dynamically fuses the semantic reference information from the hidden states of two adjacent LSTMs, the "visual sentinel" vectors of the Language LSTM and currently word embeddings, which has the potential to boost accurate captions by providing more effective textual information.

The remainder of this article is organized as follows. Some previous related studies on image captioning are introduced in Section II. Next, the overall structure and details of the proposed RASG model are documented in Section III. Then, the experimental results and analysis are given in Section IV. Finally, we conclude this study and future works are outlined in Section V.

II. Related Work

In this section, we review the existing image captioning methods in the natural image and the remote sensing fields.

A. Natural Image Captioning

The natural image captioning is a very fascinating research fields, which integrates the visual content understanding and the natural language processing (NLP) techniques to generate the descriptions for the natural images. Existing natural image captioning models can be roughly divided into three main categories, including template-based methods, retrieval-based methods, and encoder–decoder methods.

In recent years, many researchers are interested in encoder–decoder framework for the natural image captioning, where the image is first encoded into a visual vector by a CNN, and then feed it into decoder part for the generation process, such as GRU [41] or LSTM [42]. A multimodal RNN (m-RNN) [10] was proposed for generating the natural language sentence, which concatenated the image feature vector from a CNN with current word embedding vector in a multimodal layer, and then RNN was used to decode multimodal vector at every iteration. Show and tell (ST) network is a typical encoder–decoder structure method proposed by Vinyals *et al.* [11], in which a global representation extracted by encoder is translated to words via an LSTM. To address the insufficient utilization of the visual information problem in ST model, show, attend, and tell (SAT) model [24] implemented a soft/hard attention mechanism to describe the natural image content with the comprehensive sentences. The "soft" pooling aims to average with attentive weights, and the "hard" pooling only selects the most probably one-grid feature vector in the whole spatial image features. Following this work, Lu *et al.* [25] constructed a "visual sentinel" concept, namely adaptive attention mechanism. Over the generation process, the method focuses adaptively on relevant visual vectors weighted on the feature maps or nonvisual features computed from the LSTM memory vectors, which is controlled by a sentinel gate. Different from aforementioned attentive frameworks and the concatenated encoder and decoder, Anderson *et al.* [26] proposed a bottom-up and top-down attention mechanism. The bottom-up features are the regional outputs of the popular object detection network Faster R-CNN [43], which is skilled in grounding in salient image regions including objects. The decoding phase is constructed by stacking two basic LSTM layers, where the Top-Down Attention LSTM is responsible for extracting a time-varying vector from the bottom-up features, and the Language LSTM translates the visual information as the sentence. In addition, the authors regarded that their model could be used in the VQA task. For further alleviating the limited capacity of multimodal reasoning for image captioning, Huang *et al.* [27] employed an attention on attention (AOA) architecture to update relations in attentive weight by computing implicit similarity between the attention result and the query. In their approach, an AOA encoder is paired with an AOA decoder. The AOA encoder enriches image representation, and the AOA decoder explores feature interaction between image content and sentence, which is capable of multimodal reasoning. Similarly, Pan *et al.* [28] introduced a novel X-Linear attention block guided by the query, in which the attention operator was to capture higher and even infinity order interactions between encoding and decoding layers. This focus on the feature enhancement of visual feature and multimodal feature offers novel architectural modeling capabilities for image captioning.

B. Remote Sensing Image Captioning

With the development of large annotated remote sensing image datasets, e.g., Sydney-Captions, UCM-Captions, and RSICD, more and more neural network models have been proposed for the remote sensing image captioning. The multimodal encoder–decoder method proposed by Qu *et al.* [8] was the earliest work for the remote sensing image captioning, in which the feature representations of an image were derived by CNNs and then RNN or LSTM estimated the probability distribution of the next word in the condition of given previous

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words. To solve the issues of small-scale datasets of Sydney-Captions and UCM-Captions, Zhang *et al.* [29] introduced a category-aware feature for guiding language model, which included LSTM to preserve the high-level semantics of image context to improve the sentence quality. Later, inspired by the (attentive) encoder–decoder framework in the field of natural image captioning, Lu *et al.* [9] integrated different feature extractors with RNN or LSTM to generate a sentence, which included handcrafted features and convolutional features. Another contribution of this article is that they describe a large-scale remote sensing image dataset, named as RSICD. Considering the advantage of the high-level semantic information, Zhang *et al.* [30] put forward a novel attribute attention mechanism to guide the LSTM generating sentences. The method makes the decoder focus selectively on key image regions which are most relevant to the different visual attributes of input images. This attribute attention module has huge improvements on the remote sensing image caption generation. Following this work, the LAM [31] extended the attribute attention mechanism in a respect of the calculation of attention masks, in which the label information of input images was generated and combined with visual and text information to produce label-guided image features. To leverage the potential of attention mechanism, Li *et al.* [32] presented the MLA model, which combined the attentive visual features and semantic features to apply another attention to automatically construct a contextual layout. Conditioned on the contextual layout, their generator synthesizes an image matching with the text description. It appears that the MLA model has a better performance in remote sensing image captioning. Zhao *et al.* [33] developed a novel-structured attention mechanism which utilizes the segmentation proposals and spatial features to learn the structured features. The structured features can guide the decoder to accurately focus on the highly structured semantic contents during the captioning and image semantic segmentation. To make better use of the ground-truth caption, Sumbul *et al.* [35] proposed a summarization-driven image captioning method, which summarizes ground-truth captions into single caption and applies an adaptive weighting strategy to integrate the summarized captions with standard captions for improving the generalization capability of their designed approach. In particular, the proposed VRTMM [38] encoded better representation for given remote sensing image by a variational autoencoder module and utilized the transformer model [23] with reinforcement learning to model long-term dependencies in a sequence. The experiment results on RSICD dataset achieve the favorable accuracy.

However, the aforementioned methods either failed to exploit the characteristic of the remote sensing images, or trained not in end-to-end pattern. With considering the ambiguous, abstractive, and complex properties of the remote sensing images, we will present a novel approach for translating images as the natural language, which employs a recurrent attention network on competitive visual features and integrates different semantics information through a semantic gate. Moreover, the proposed approach could process sequence prediction more flexibly.

III. Proposed Approach

Our method follows the attentive encoder-decoder approach and tries to improve the performance of feature extractor and language modeler. In this section, we elaborate on our proposed approach, which is split into three distinct components including feature extractor, recurrent attention, and semantic gate. The framework of our RASG model is illustrated in Fig. 2.

1) Feature Extractor: The competitive features are obtained by applying concatenation operation on original CNN features and the global feature of the multiscale features.

2) Recurrent Attention: Taking the guidance of the past attention result and the contribution of nonvisual features into consideration, the recurrent attention can generate the comprehensive and accurate contextual vectors associated with the remote sensing images.

3) Semantic Gate: Utilize the different semantic representations to construct the current word state to generate description.

A. Notations

Before presenting our method, we first introduce the notations used in this article, as shown in the Nomenclature.

B. Feature Extractor

The encoded features V_{enc} contain two different features: the original CNN features and the global feature of multiscale features, where the original CNN features contain the spatial information of objects, and the global feature of multiscale features contains the multiscale, relational, and global information about input images. In this article, we select the pretrained VGG16 [44] to process the remote sensing image and employ the result of the last convolution layer as the original CNN features. Then, we get the feature representation $V_{\text{CNN}} = \{v_1, v_2, \ldots, v_k\}$, $v_k \in R^D$, where k is the counts of image location features and D represents the dimension of each feature.

We first get multiscale features V_{ESP} using an ESP module, where the dilated convolution filters with different dilation rates learn weights of different receptive fields. The structure of the ESP module is shown in Fig. 3. The ESP module takes feature maps V_{CNN} as input, and the pointwise convolution in the ESP module applies a 1×1 convolution to compress the high-dimensional spatial features into a low-dimensional space dimension (D/K). Then the low-dimensional feature maps are simultaneously across K parallel dilated convolutional branches with different dilation rates. The obtained feature maps are added hierarchically before concatenating them to address the checkerboard or gridding artifacts in each dilated convolution branch. The input and output feature maps of the ESP module are combined using an element-wise sum as the final output, denoted as $V_{\text{ESP}} = \{v'_1, v'_2, \ldots, v'_k\}$. In our RASG model, K is set to 4, the size of dilated convolutional kernels is 3×3, and the dilation rates are set to 2^n, $n = \{1, \ldots, 4\}$. Finally, we design an efficient way to extract a global feature applying an adaptive-avg layer on

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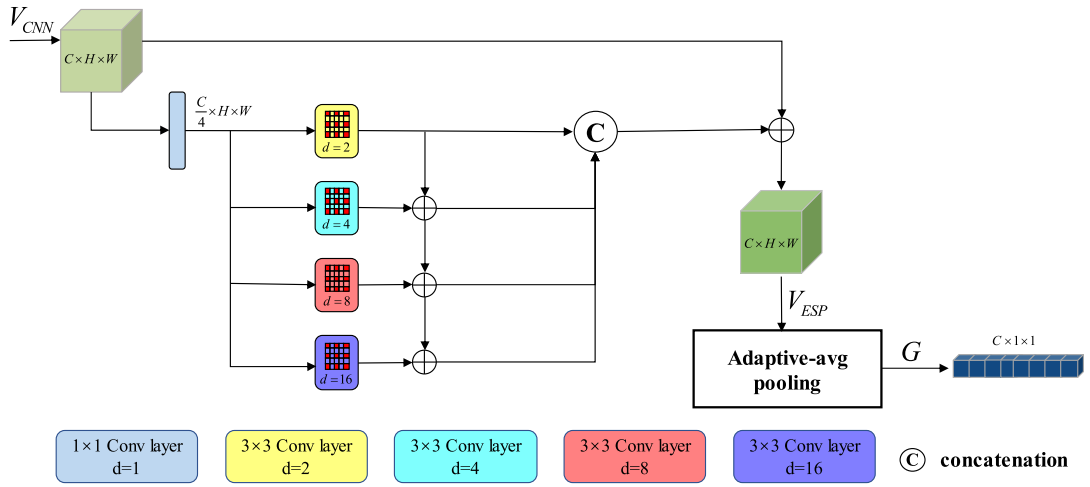


Fig. 3. Architecture for the global feature G. The multiscale features V_{ESP} are obtained via ESP module which has four steps: reduce feature dimension, split-flow, dilated convolution processes, and merge.

V_{ESP}. Therefore, each remote sensing image can be finally represented as encoded vectors $V_{\mathrm{enc}} = \{G, v_1, v_2, \ldots, v_k\}$, where G represents the specific global feature.

C. Novel Recurrent Attention Model

Different from bottom-up and top-down attention model [26] that two basic LSTMs with a top-down attention were used for image description, we design the two LSTM extensions in the decoder part, namely Attention LSTM and Language LSTM. Our novel decoder module is illustrated in Fig. 2, where the Attention LSTM and Language LSTM are designed for the recurrent attention and language modeling. The input vector to the Attention LSTM layer at each time step consists of the mean-pooled image feature $\bar{v}$, the previous hidden state h_{t-1}^2 from the Language LSTM layer and the current word embedding ω_t, given by

$$x_t^1 = \left[\bar{v}, h_{t-1}^2, \omega_t\right] \tag{1}$$

where $\bar{v} = (1/k)\sum_{i=1}^{k} v_i$ is computed from V_{CNN}, and ω_t is an one-hot word vector.

These inputs make the image feature compatible with the decoder. Our extend LSTM is designed with a concept of "visual sentinel" [25], which is dependent on what the Attention LSTM already knows. At each iteration, the hidden state h_t^1 and "visual sentinel" vector s_t^1 of the Attention LSTM could be expressed as

$$\begin{aligned} h_t^1 &= \mathrm{LSTM}_1\left(x_t^1, h_{t-1}^1\right) \\ g_t^1 &= \sigma\left(W_x^1 x_t^1 + W_h^1 h_{t-1}^1\right) \\ s_t^1 &= g_t^1 \odot \tanh\left(m_t^1\right) \end{aligned} \tag{2}$$

where h_{t-1}^1 is the hidden state of the Attention LSTM at the $t-1$ time, the parameters of W_x^1 and W_h^1 can be learned, and the gate of the Attention LSTM is g_t^1 which is applied on the memory cell vector m_t^1.

In general, the attentive result att_t just focuses on attractive objects for the image [24], [26], which plays an important role in decoder module. However, the attentive result should have contextual dependencies and provide visual/semantic evidence for later time steps. Thus we propose the recurrent attention module. For modeling, the contextual dependencies between adjacent attentive results, we treat att_{t-1} as previous attended result and integrate it with the hidden state h_t^1 to define a new hidden state H_t. Formally, let $H_t = h_t^1 \circ \mathrm{att}_{t-1}$. Then H_t is employed to distinguish adaptively the importance of the encoded features and nonvisual semantics (i.e., "visual sentinel" vector s_t^1). If language generation make the visual information unnecessary when generating words, larger weights will be assigned for nonvisual representation.

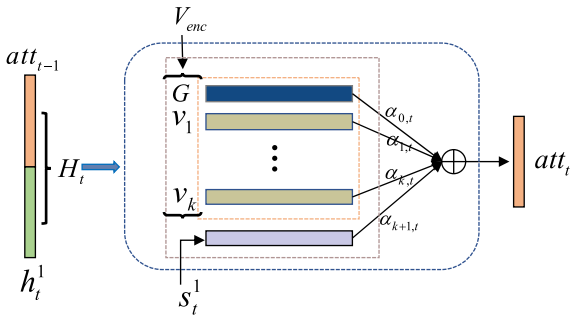


Fig. 4. Architecture of the recurrent attention module. Given new hidden state H_t extracted via the concatenation of previous attention vector att_{t-1} and the current hidden state h_t^1 of the Attention LSTM, the recurrent attention is conducted over the encoded features V_{enc} and the "visual sentinel" vector s_t^1.

As illustrated in Fig. 4, given the encoded visual features V_{enc}, the new hidden state H_t and s_t^1 from the Attention LSTM, att_t with the recurrent module is gotten. The current attentive vector att_t is defined as follows:

$$\mathrm{att}_t = f_{\mathrm{att}}\left(V_{\mathrm{enc}}, H_t, s_t^1\right) \tag{3}$$

where $f_{\mathrm{att}}(\cdot)$ indicates the recurrent attention function. At each time step t, we generate a attention weight α_t for each element among V_{enc} and s_t^1 as follows:

$$\begin{aligned} u_t &= \omega_u^T \tanh\left(W_{\mathrm{vu}} V_{\mathrm{enc}} + (W_{\mathrm{Hu}} H_t)\mathrm{I}^T\right) \\ \alpha_t &= \mathrm{softmax}\left(\left[u_t; \omega_h^T \tanh\left(W_s^1 s_t^1 + W_z^1 h_t^1\right)\right]\right) \end{aligned} \tag{4}$$

where W_{vu}, W_{Hu} and ω_u^T are trainable parameters in $f_{\mathrm{att}}(\cdot)$, $\mathrm{I} \in R^{k+1}$ is a vector with all elements set to 1, the vector u_t contains attention scores distributed in V_{enc}, W_s^1 and W_z^1 are parameters that transform each "visual sentinel" element

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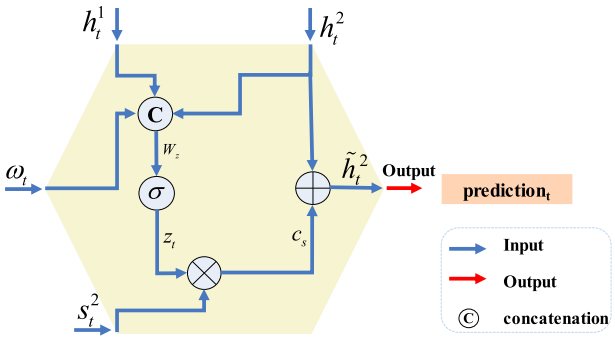


Fig. 5. Illustration of the semantic gate. The h_t^2, h_t^1, w_t, and s_t^2 are fed into the designed gate. The output of gate is denoted as the final word state representation at current time.

and hidden state into a latent space, α_t is a $k+2$ dimensional vector and satisfies the constraint of $\sum_{i=0}^{k+1} \alpha_{i,t} = 1$. To obtain the attention vector att_t, we apply weights to combine each element among V_{enc} and s_t^1 at each time step

$$\text{att}_t = \alpha_{0,t} G + \sum_{i=1}^{k} \alpha_{i,t} v_j + \alpha_{k+1,t} s_t^1. \tag{5}$$

D. Semantic Gate for Deliberate Captions

Most existing language generators are based on LSTM, which took the hidden state from a LSTM as current word state h_t for repeated word prediction. Due to the simplicity of word states at every time point, their system might easily result in an undesirable inference. Inspired by [25] and [26], the word embeddings, hidden states and "visual sentinel" vectors of the LSTM include additional semantic information, thus the semantic gate is proposed for effective textual information learning automatically.

The input of the Language LSTM consists of the attentive result att_t and the hidden state h_t^1 from the Attention LSTM, given by

$$x_t^2 = \left[\text{att}_t, h_t^1\right]. \tag{6}$$

For the Language LSTM, we adopt similarly an extended LSTM with the concept of "visual sentinel" [25]. The current hidden state h_t^2 and the "visual sentinel" vector s_t^2 from the Language LSTM are defined respectively as

$$\begin{aligned} h_t^2 &= \text{LSTM}_2\left(x_t^2, h_{t-1}^2\right) \\ g_t^2 &= \sigma\left(W_x^2 x_t^2 + W_h^2 h_{t-1}^2\right) \\ s_t^2 &= g_t^2 \odot \tanh\left(m_t^2\right) \end{aligned} \tag{7}$$

where h_{t-1}^2 is the hidden state of the Language LSTM at previous time step, the parameters of W_x^2 and W_h^2 can be learned, m_t^2 represents the memory state of the Language LSTM, and the gate of the Language LSTM is g_t^2 which is applied on the memory cell vector m_t^2.

To make the most use of the underlying semantic information in decoder module, the semantic gate structure integrates different types of semantic representations for predicting word, as shown in Fig. 5. The inputs are current word embedding ω_t, adjacent layer's hidden states $\{h_t^1, h_t^2\}$, as well as the "visual sentinel" vector s_t^2 of the Language LSTM. We first employ concatenation on $\{\omega_t, h_t^1, h_t^2\}$, and then perform an element-wise multiplication of the intermediate semantic vector z_t with s_t^2, which may be thought of as calculating the relevance of semantic information associated with the word states at each time step

$$\begin{aligned} c_s &= z_t \odot \tanh\left(s_t^2\right) \\ z_t &= \sigma\left(W_z\left[w_t; h_t^1; h_t^2\right]\right) \end{aligned} \tag{8}$$

where W_z is trainable parameter. After getting the supplementary semantic representation c_s, we integrate this representation and the original word state h_t^2 with summation, which can be formalized as follows:

$$\tilde{h}_t^2 = h_t^2 + c_s \tag{9}$$

where $\tilde{h}_t^2$ represents the word semantics by a comprehensive way and guarantees the performance of word prediction. Finally, the probability p_t over the vocabulary of words is computed by

$$p_t = \text{softmax}\left(\tilde{h}_t^2\right). \tag{10}$$

E. Objective Function

To optimize the proposed framework, we use the cross entropy loss and the discriminative loss to construct our loss function.

1) Cross Entropy Loss: Cross entropy loss is a generic objective in the remote sensing captioning, which aims to minimize a weighted sum of the negative conditional log likelihood. Given a remote sensing image I, the image caption model generates a sentence $S = \{s_0, s_1, \ldots, s_{N-1}\}$ with N words. The cross entropy loss function of our network is defined as following:

$$\begin{aligned} \text{loss}_{\text{ce}} &= -\log p(S|I;\theta) \\ &= -\sum_{t=0}^{N-1} \log\left(p_\theta(s_t|I, s_{0:t-1})\right) \end{aligned} \tag{11}$$

where θ denotes all learned parameters in the captioning model.

2) Discriminative Loss: To make the hidden states of the Attention LSTM more semantically correlated, we leverage discriminative supervision by calculating loss on the hidden states of the Attention LSTM. The discriminative supervision has been proven to be effective on reviewer modules [45] and integrated into image captioning model in an end-to-end manner.

In our RASG model, taking a ground-truth caption $Y = \{y_0, y_1, \ldots, y_{N-1}\}$ for example, a max-pooling across the hidden states of the Attention LSTM over all the time steps is used to extract the most salient signal, so a multilabel margin loss is added as discriminative supervision. Let $S_{\max}$ be the highest score after the max-pooling layer, and W be the set of all words in the ground-truth sentence. The discriminative loss is defined as

$$\text{loss}_{\text{dis}} = \frac{1}{Z} \sum_{j \in W} \max\left(0, 1 - \left(y_j - s_{\max}\right)\right) \tag{12}$$

where the $(s_{\max}, y_j)$ pairs are denoted by Z.

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The loss function of our model is the sum of cross entropy loss and the discriminative loss. It is shown as follows:

$$\text{loss} = \text{loss}_{\text{ce}} + \lambda \cdot \text{loss}_{\text{dis}} \tag{13}$$

where λ is a hyperparameter, which is set as 10 in this article.

IV. Experiment Results

In this section, we will introduce the three well-known benchmark datasets and the adopted evaluation metrics. We also compare our results with several state-of-the-art methods. Then, we will give experimental results and detailed analysis of these models.

A. Datasets

1) Sydney-Captions: Based on the Sydney Dataset [46], Sydney-Captions [8] includes seven land cover categories and has total 613 remote sensing images, where the size of each image is 500 × 500 pixels. In particular, each image is associated with five human-annotated sentences which have the same meaning.

2) UCM-Captions: UCM-Captions Dataset [8] is based on the UC Merced Land Use data [47], which contains 2100 images with a size of 256 × 256 pixels and 21 land cover categories. There are five available sentences per image and each sentence has different length.

3) RSICD: Compared to the natural image description on MSCOCO [48] and Flickr30K [49], the remote sensing image captioning lacks a very large-scale benchmark dataset. The RSICD provided by Lu *et al.* [9] is originated in Google earth, with a collection of 10 921 remote sensing images and five descriptive descriptions for each image. Each remote sensing image scene is with a size of 224 × 224. Furthermore, the dataset provides comprehensively 30 land cover categories to help researchers devise novel approaches for the remote sensing image captioning.

B. Experimental Setup

1) Preprocessing: Similar with existing remote sensing image captioning [30], the publicly available splits are adapted in our experiment. The ratio of training set, validation set, and testing set are 80%, 10%, and 10% respectively. We have shuffled each train dataset to avoid overfitting or nonconvergence in training phase. And the size of each input image is set to 224 × 224 pixels. Then, after the VGG16 pipeline, the extracted feature maps (i.e., before the pooling layer) has size 14 × 14 × 512.

To prepare input for our remote sensing image captioning system, we first strip out individual words from the description corpus, and then convert all the words to lowercase by employing word_tokenize operation in the NLTK toolbox2. Each sentence starts with $\langle start \rangle$ and ends with $\langle end \rangle$. Specifically, all the words appeared in the Sydney-Captions and UCM-Captions are used to build vocabulary. The two vocabularies have 238 and 342 unique words, respectively. RSICD vocabulary is with the size of 1806 by cleaning these words which appear less than five times. Each vocabulary contains special masks (e.g., $\langle start \rangle$, $\langle end \rangle$), and each input word is represented as an one-hot vector with the dimension of its corresponding vocabulary size.

2) Model Implementation: Our model is implemented in PyTorch. To extract the enhanced features, the original CNN features with size of 14 × 14 × 512 are the "conv5_3" features of VGG16 pretrained on the ImageNet dataset [50]. Specifically, these features are fed to the ESP module for a global feature of multiscale features. By concatenating the two features, we get the competitive features with the dimension of $(14 \times 14 + 1) \times 512$.

For decoder module, we set the word embedding dimension as 512. The hidden state size of two extended LSTMs is 1024. The hidden size in recurrent attention calculation is set as 512. The output sizes of the semantic gated in the word prediction layer is set as 512.

The training steps are performed by an Adamax optimizer with alpha 0.8 and beta 0.999. The initial learning rates of encoder module and decoder module are set to 1×10^{-5} and 5×10^{-4}, respectively. The decoder's initial learning rate will be annealed by a factor of 0.8 when eight epochs have not improved, and the encoder's initial learning rate will be fine-tuned after five epochs. We train our model using minibatch with size 64. The training process will stop early when the validation CIDEr score has not improved for 20 consecutive epochs. Our model can be trained in less than 6 h on single Nvidia GeForce GTX 1080Ti GPU. We use a beam size of 3 in the sampling process.

C. Evaluation Metrics

To comprehensively compare our RASG model with several existing state-of-the-art approaches, we illustrate the remote sensing image captioning performance by nine metrics of the experimental results similar to [31]: BLEU-n [51], METEOR [52], ROUGE-L [53], CIDEr [54], SPICE [55], and S_m [31]. Early metrics on the machine translation have been adopted in image description, such as BLEU-n, METEOR, ROUGE-L, CIDEr, and SPICE. Specifically, CIDEr reflects consistency and smoothness among sufficient reference sentences, and SPICE focuses on the semantic scene graphs of objects, attributes, and relationships between the candidate and reference captions. BLEU-n (the range of n is from 1 to 4) represents n-gram precision between the reference sentences and generated sentence, which captures the lexical and textual consistency between sentences. ROUGE-L only calculates recall ratio of the shared subsequence with maximum length between the generated sentence and the ground-truth sentences. However, BLEU-n and ROUGE-L cannot measure the semantic similarity. With the help of the WordNet database, METEOR can capture the semantic aspects and compute sentence level similarity scores. S_m is an overall measure that integrates most metrics in the evaluation. Formally, it is defined as follows:

$$S_m = \frac{1}{5}(B_4 + M + R_L + C + S) \tag{14}$$

where B_4, M, R_L, C, and S represent the metric BLEU-4, METEOR, ROUGE-L, CIDEr, and SPICE, respectively.

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TABLE I
COMPARISON SCORES OF OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON SYDNEY-CAPTIONS DATASET [8]

Methods	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
SAT [24]	0.7905	0.7020	0.6232	0.5477	0.3925	0.7206	2.2013	-	-
FC-Att [30]	0.8076	0.7160	0.6276	0.5544	0.4099	0.7114	2.2033	-	-
SM-Att [30]	0.8143	0.7351	0.6586	0.5806	0.4111	0.7195	2.3021	-	-
Ada-Att [25]	0.7890	0.7112	0.6417	0.5790	0.3877	0.7082	2.4611	0.4265	0.9125
Up-Down [26]	0.8042	0.7330	0.6718	**0.6167**	0.4125	**0.7418**	**2.7281**	**0.4567**	**0.9911**
LAM+ [31]	0.7405	0.6550	0.5904	0.5304	0.3689	0.6814	2.3519	0.4308	0.8671
MLA [32]	**0.8152**	**0.7444**	**0.6755**	0.6139	**0.4560**	0.7062	1.9924	0.3890	0.8449
VRTMM [38]	0.7443	0.6723	0.6172	0.5699	0.3748	0.6698	2.5285	-	-
Ours	0.8000	0.7217	0.6531	0.5909	0.3908	0.7218	2.6311	0.4301	0.9529

* The "-" means that the scores are not reported in the reference papers.

TABLE II
COMPARISON SCORES OF OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON UCM-CAPTIONS DATASET [8]

Methods	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
SAT [24]	0.7993	0.7355	0.6790	0.6244	0.4174	0.7441	3.0038	-	-
FC-Att [30]	0.8135	0.7502	0.6849	0.6352	0.4173	0.7504	2.9958	-	-
SM-Att [30]	0.8154	0.7575	0.6936	0.6458	0.4240	0.7632	3.1864	-	-
Ada-Att [25]	0.8260	0.7654	0.7163	0.6720	0.4564	0.7918	3.2788	0.4782	1.1354
Up-Down [26]	0.8394	0.7792	0.7305	0.6862	0.4535	0.7978	3.3975	**0.4940**	1.1658
LAM+ [31]	0.8195	0.7764	0.7485	0.7161	0.4837	0.7908	3.6171	0.5024	1.2219
MLA [32]	0.8406	0.7803	0.7333	0.6916	**0.5330**	**0.8196**	3.1193	0.4601	1.1247
VRTMM [38]	0.8394	0.7785	0.7283	0.6828	0.4527	0.8026	**3.4948**	-	-
Ours	**0.8518**	**0.7925**	**0.7432**	**0.6976**	0.4571	0.8072	3.3887	0.4891	**1.1679**

* The "-" means that the scores are not reported in the reference papers.

D. Baselines

In this section, we evaluate our model by comparing with several state-of-the-art attentive encoder-decoder approaches, which will be reviewed briefly as below.

1) SAT [24]: SAT extracts visual features through a pretrained CNN and applies an attention mechanism to obtain context vector, then translates to natural language using a single basic LSTM.

2) FC-Att/SM-Att [30]: This model is an improved attribute attention mechanism for remote sensing image captioning, the high-level attributes that are constructed by the final fully connected layer or soft-max layer of a pretrained CNN. A basic LSTM is used for predicting words one by one.

3) Ada-Att [25]: The model employs a "visual sentinel" vector to determine whether to attend the visual features or linguistic knowledge to predict next word by the basic LSTM.

4) Up-Down [26]: Up-Down stacks two basic LSTM layers for visual attention and language decoding. Up-Down is the most likely approach with ours.

5) LAM+ [31]: LAM integrates label information with an attention mechanism for exploiting redundant image features of the remote sensing image. Then a basic LSTM is used to preserve the attentive maps for better sentence representations. It is should be noted that the preprocessing of experimental datasets is different from other baselines, so we use special notation "+" to mark it.

6) MLA [32]: MLA integrates a pretrained CNN and a basic LSTM with the MLA for the remote sensing image captioning, which can automatically choose whether to focus on the image or the language.

7) VRTMM [38]: VRTMM is a two-stage model for the remote sensing image captioning task. The representation for given image is obtained by the pretrained variational autoencoder module, and the transformer network is used to generate better description.

E. Evaluation Results and Analysis

1) Quantitative Comparison: For the fairness of the experimental comparison, the VGG16 is employed to construct the encoder module of comparison methods. Tables I–III shows the performances of our model and other attention-based models on Sydney-Captions dataset, UCM-Captions dataset and RSICD dataset, respectively. It is not hard to find that our proposed method achieves higher scores on all metrics than the contrast models. This is because the introducing of encoded competitive features is to make the model comprehensively understand the content of a remote sensing image. With the high-level contextual vectors obtained by the recurrent attention, better sentence semantic representation can be easily constructed. Moreover, a semantic gate can restrict effectively the generation to understand which objects corresponding to which words with integrating different semantics. Thus, our RASG model has the capability to narrow down the gap between texts and images.

As we can see, the "SAT" gets really the lowest scores in Tables I–III. This infers that only using visual features is

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TABLE III
COMPARISON SCORES OF OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON RSICD DATASET [9]

Methods	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
SAT [24]	0.7336	0.6129	0.5190	0.4402	0.3549	0.6419	2.2486	-	-
FC-Att [30]	0.7459	0.6250	0.5338	0.4574	0.3395	0.6333	2.3664	-	-
SM-Att [30]	0.7571	0.6336	0.5385	0.4612	0.3513	0.6458	2.3563	-	-
Ada-Att [25]	0.7128	0.5941	0.5024	0.4290	0.3426	0.6240	2.3061	0.4298	0.8263
Up-Down [26]	0.7708	0.6542	0.5614	0.4848	0.3592	0.6580	2.6237	0.4658	0.9183
LAM+ [31]	0.6753	0.5537	0.4686	0.4026	0.3254	0.5823	2.585	0.4636	0.8717
MLA [32]	0.7725	0.6290	0.5328	0.4608	0.4471	0.6910	2.3637	0.4315	0.8789
VRTMM [38]	**0.7813**	**0.6721**	0.5645	**0.5123**	**0.3737**	**0.6713**	2.7150	-	-
Ours	0.7729	0.6651	**0.5782**	0.5062	0.3626	0.6691	**2.7549**	**0.4719**	**0.9529**

* The "-" means that the scores are not reported in the reference papers.

not enough for the caption generation. Since the "Ada-Att" model discovers adaptively not only visual information but also the linguistic information from the memory of a basic LSTM, it outperforms "SAT" model on three public datasets. It is worth mentioning that the design concepts of attention mechanism in the "MLA" are similar to the "Ada-Att," but the "MLA" can makes the result better than the "Ada-Att." Since the "MLA" adds attention to the historical semantic information.

In Tables I–III, the "FC-Att," "SM-Att," and "LAM+" belong to the attribute-based models, in which the attentive regional features and high-level image features (attribute vectors) are used to generate sentence. Tables I–III clearly reports that the attribute-based models show better performance than attention-based models. This is because attribute vectors contain label information which can stand for the content of images concisely and accurately. Among attention-based models, the results of the "MLA" are near to the "FC-Att" and "SM-Att." By adding the attribute vectors or generated semantic information, the models are fed with more effective multimodal features when generating words, which will improve the scores. It is noteworthy that an attribute generator is specially trained in the "LAM+" model and not in "FC-Att" and "SM-Att." Thus, "LAM+" is better than both "FC-Att" and "SM-Att." In Tables I and II, the "MLA" model performs slightly better than our models on some evaluation metrics. In particular, the results of the "MLA" on Meteor that can capture the semantic aspects scores outperform all of the other approaches because of the usage of the MLA mechanism. However, the performance of the "MLA" model on RSICD dataset is not competitive.

The models applied the double-layer LSTM decoder (i.e., "Up-Down," Ours) can obtain higher scores than almost all the single-layer LSTM decoder structure correspondingly. The phenomenon demonstrates that the bilayer LSTM structure benefits from encoding remote sensing images and decoding for natural language separately, rather than using preliminary context information at decoder stage. We observe that the "Up-Down" model is a strong baseline for image captioning. Though the results of the "Up-Down" are obtained with a better performance on Sydney-Captions dataset which is the smallest dataset, the performance of the model on UCM-Captions and RSICD datasets is inferior to our method. The model (i.e., the "Up-Down," Ours) trained under a large dataset is more robust. It follows that our results are competitive and possibly better.

Different from the LSTM-based decoder, the "VRTMM" utilizes the transformer with six subdecoders as the decoder, which models the sequence dependency with the attention mechanism in parallel. The transformer has been proven superior to RNN or LSTM in natural image captioning. In addition, we note that the encoder of the "VRTMM" is pretrained with multitask for encoding images to obtain better representation in the remote sensing filed. Since the two-stage model can usually boost the performance, the scores of the "VRTMM" on RSICD exceed slightly our model. In Tables I and II, the scores of our models are higher than the "VRTMM," which indicates the former learns a better language model than the latter under the supervision of small datasets.

We note that other approaches only use the original CNN features to propagate the whole scene information, but they miss certain targets which are too small and contain too little contextual information. Thus, the encoded competitive features are adopted in our RASG model instead of the conventional image features. Obviously, the generation performance improves compared with "Up-Down." Here, we conduct an ablation analysis in Table VII. It shows the importance role of the encoded competitive features obtained by our feature extractor. The encoded features are one input of recurrent attention. In addition, we extend a basic LSTM to produce new features ("visual sentinel" vectors) as another input of recurrent attention, which contain existed semantic representations. Simultaneously, a new current hidden state is constructed by taking contextual relevance into account. In this way, the effective attentive maps obtained with recurrent attention can produce nonvisual sensitive features. Following the Language LSTM guided by the attention vectors, the semantic gate is applied for combining more semantics in the decoder stage. This gate is beneficial for exploiting the more strengthened word state to describe the content of images as time flows.

Besides, it is observable that the results obtained by models on Sydney-Captions and UCM-Captions are higher, while models experimented on RSICD show the lowest scores. This phenomenon demonstrates that the results of these image

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TABLE IV
EXPERIMENTAL RESULTS ABOUT EFFECT OF DIFFERENT CNNS FOR OUR MODEL ON SYDNEY-CAPTIONS DATASET [8]

Methods	CNN	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
Up-Down [26]	VGG11	0.7954	0.7135	0.6442	0.5844	0.3978	0.7294	2.6072	0.4490	0.9536
	VGG16	**0.8042**	**0.7330**	**0.6718**	**0.6167**	**0.4125**	**0.7418**	**2.7281**	**0.4567**	**0.9911**
	VGG19	0.7768	0.7054	0.6457	0.5938	0.3995	0.7230	2.6373	0.4333	0.9574
Ours	VGG11	0.7929	0.7153	0.6506	0.5929	0.4015	0.7251	**2.6642**	0.4425	0.9652
	VGG16	0.8000	0.7217	0.6531	0.5909	0.3908	0.7218	2.6311	0.4301	0.9529
	VGG19	**0.8078**	**0.7355**	**0.6762**	**0.6241**	**0.4097**	**0.7409**	2.6618	**0.4681**	**0.9809**

TABLE V
EXPERIMENTAL RESULTS ABOUT EFFECT OF DIFFERENT CNNS FOR OUR MODEL ON UCM-CAPTIONS DATASET [8]

Methods	CNN	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
Up-Down [26]	VGG11	0.8330	0.7644	0.7101	0.6630	0.4467	0.7923	**3.4138**	0.4879	1.1607
	VGG16	**0.8394**	**0.7792**	**0.7305**	**0.6862**	**0.4535**	**0.7978**	3.3975	**0.4940**	**1.1658**
	VGG19	0.8213	0.7650	0.7184	0.6768	0.4377	0.7809	3.3282	0.4790	1.1405
Ours	VGG11	0.8492	**0.7933**	**0.7477**	**0.7053**	0.4501	0.7964	**3.4675**	0.4834	1.1806
	VGG16	**0.8518**	0.7925	0.7432	0.6976	**0.4571**	**0.8072**	3.3887	**0.4891**	**1.1679**
	VGG19	0.8308	0.7632	0.7111	0.6640	0.4391	0.7899	3.3500	0.4823	1.1451

TABLE VI
EXPERIMENTAL RESULTS ABOUT EFFECT OF DIFFERENT CNNS FOR OUR MODEL ON RSICD DATASET [9]

Methods	CNN	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
Up-Down [26]	VGG11	0.7695	**0.6614**	**0.5734**	**0.4994**	**0.3598**	**0.6654**	**2.6313**	**0.4700**	**0.9252**
	VGG16	**0.7708**	0.6542	0.5614	0.4848	0.3592	0.6580	2.6237	0.4658	0.9183
	VGG19	0.7557	0.6414	0.551	0.4756	0.3579	0.6548	2.5621	0.4636	0.9028
Ours	VGG11	0.7716	0.66	0.5694	0.4938	0.3614	0.6663	2.6634	0.4676	0.9305
	VGG16	**0.7729**	**0.6651**	**0.5782**	**0.5062**	**0.3626**	**0.6691**	**2.7549**	**0.4719**	**0.9529**
	VGG19	0.7708	0.6556	0.5636	0.4873	0.3598	0.6619	2.6394	0.4678	0.9232

caption models are sensitive to the size of dataset. As we know, compared with UCM-Captions and Sydney-Captions, RSICD contains the most remote sensing images with various descriptions, which means that a large vocabulary will be constructed for RSICD. The generated sentence will be more semantically-rich and deliberate due to the large vocabulary. However, larger scale of dataset and vocabulary may bring more difficulties for training the models. Therefore, the scores obtained by models on Sydney-Captions and UCM-Captions are higher than models on RSICD. For all that, our proposed method achieves significantly better performance than five methods (except "LAM+") on the three datasets.

Next, we evaluate the influence of adopting different variants of CNNs features for the sentence generation. The performances on Sydney-Captions, UCM-Captions, and RSICD are shown in Table IV–VI separately, combining different CNNs features learned from VGG11, VGG16, and VGG19 [44]. Clearly, the proposed method outperforms "Up-Down" model on all evaluation metrics. On Sydney-Captions dataset, "Up-Down" model based on VGG16 obtains the highest scores, and our model based on VGG19 gets the highest scores. We observe that the metrics of our model on UCM-Captions dataset are higher than the best previous results of "Up-Down" model with three CNNs (VGG11, VGG16, and VGG19). We also compare the results of our model and "Up-Down" model on RSICD dataset. In terms of overall metrics, our model based on VGG16 gets the highest scores and significantly outperforms the best result obtained by "Up-Down" model. Obviously, injecting an excellent CNN backbone with our model demonstrates much better performance than the unfiltered one. As indicated by our results, utilizing RSICD to train our model can constantly lead to better performance than Sydney-Captions and UCM-Captions which reflect the robustness of the proposed model under the supervision of the most complicated dataset RSICD.

2) Qualitative Comparison: Fig. 6 exhibits a few remote sensing image captioning results with a comparison of ground truth, "Up-Down," and our model. We can see from these examples that our model is able to capture more details of remote sensing images than "Up-Down." It implies that the encoded competitive features with a recurrent attention structure could enhance the capacity of our model which dynamically allocates weights for visual and nonvisual features in each step. Moreover, the semantic gate digs more important sematic information to generate more detailed descriptions than sentences generated by the alone hidden state of the Language LSTM.

In Fig. 6, we highlight some phrases with blue color, which is important part corresponding to the scene and content of remote sensing images. Red color is used to mark generated

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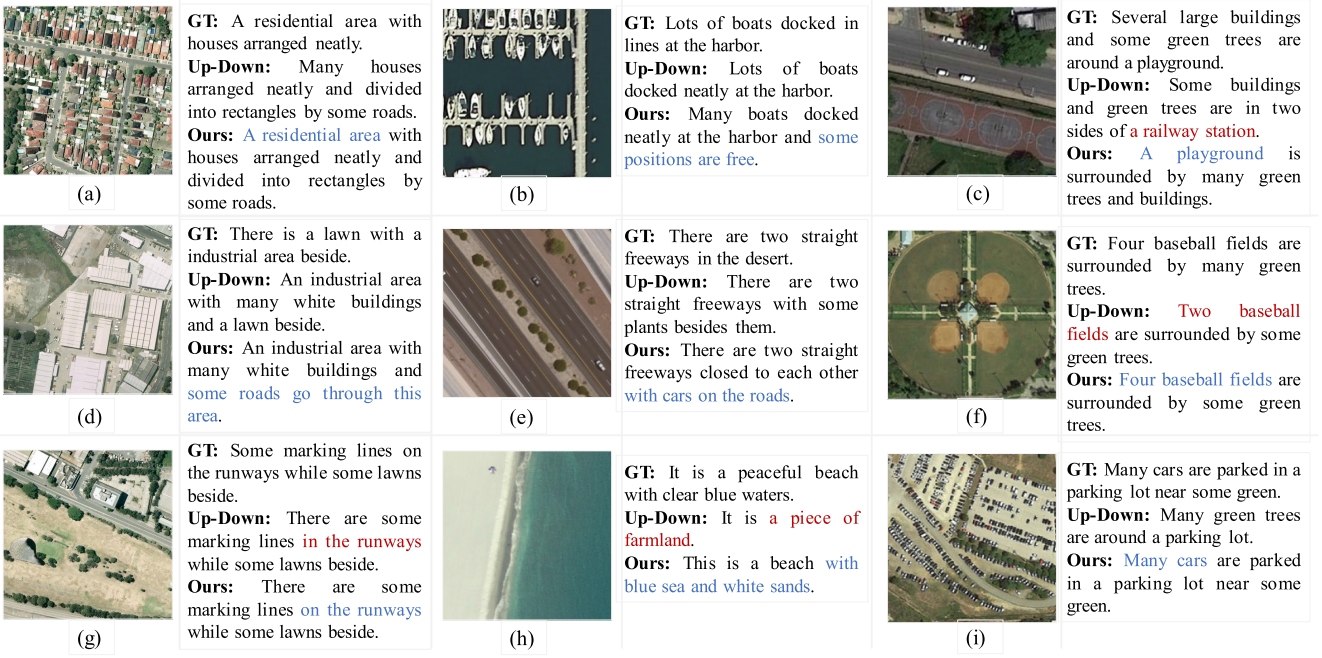


Fig. 6. (a), (d), and (g) Sydney-Captions. (b), (e), and (h) UCM-Captions. (c), (f), and (i) RSICD. The output sentences are generated by: 1) ground truth (GT): one selected ground truth sentence; 2) up-down model; and 3) our proposed model. The red words indicate mismatching in the generated images, and the blue ones are precise words with our model.

wrong words. It is shown that our model is able to capture more complex semantic content of remote sensing images and generate more accurate and complete descriptions compared to the "Up-Down" model. Fig. 6(a) is a residential area, but "Up-Down" neglects this scene information which can be described in the generated sentence by our RASG model. For Fig. 6(c), the scene "a playground" which is main element of the image is described robustly by our model and the wrong scene "a railway station" is generated by "Up-Down." Especially, the component of a beach is generated correctly by our RASG model, as shown in Fig. 6(h) image. What is more, the sea's color attribute and the sand's color attribute are also described in the generated sentence, because our model can utilize more precise visual information than "Up-Down." For Fig. 6(d) image, the roads that are not salient visual information distributed throughout the image can also be described by our model. It is similar to Fig. 6(e) image. In addition, some small objects (such as cars) that are hard to be captured appear in the sentence generated by our model. Fig. 6(i) has the same characteristic with the effectiveness of enhancing the small objects semantic understanding. What is more, some information in our generated sentence does not even exist in the ground-truth, which in fact appears in the image like the "free positions" in Fig. 6(b) image. For Fig. 6(f) image, "Up-Down" may succeed in understanding the general context of the image, but fails to predict the number of baseball filed, which indicates that the model is not sensitive to numbers. It is worth mentioning that our RASG model can modify generation process to yield more accurate captions. An example of this phenomenon is shown in Fig. 6(g), in which our model modifies successfully caption with "on the runways." All the instances prove that our RASG model is capable of achieving good performance in the remote sensing image captioning.

F. Ablation Experiments

In Section IV-E, we have proven that our model has a big improvement for the remote sensing image captioning. However, the importance of certain components in our proposed model is still unclear. Thus, we conduct an ablation study to examine the effectiveness of each component on RSICD dataset.

1) Bottom-up and Top-down attention is used for the remote sensing image captioning (Up-Down). The Faster-RCNN network is abandoned. The image features are extracted from the "conv5_3" layer of the VGG16.
2) Encoded features (EF). Only the more comprehensive feature extractor network is used.
3) Encoded features and recurrent attention (EF + RA). The recurrent attention network is applied on encoded features and nonvisual information.
4) Encoded features with recurrent attention and semantic gate (FULL). Our full model.

1) Quantitative Comparison: To objectively demonstrate the superiority of each component in our designed model, the ablation models are based on the popular VGG16 in the encoding stage. Table VII shows the performance of the ablation models. From the results, our proposed models perform better when consider the encoded competitive features, the recurrent attention and the semantic gate. First, the EF model improves the performance compared with Up-Down model. Generally, the multiscale features consider about the small-scale object information from large effective receptive

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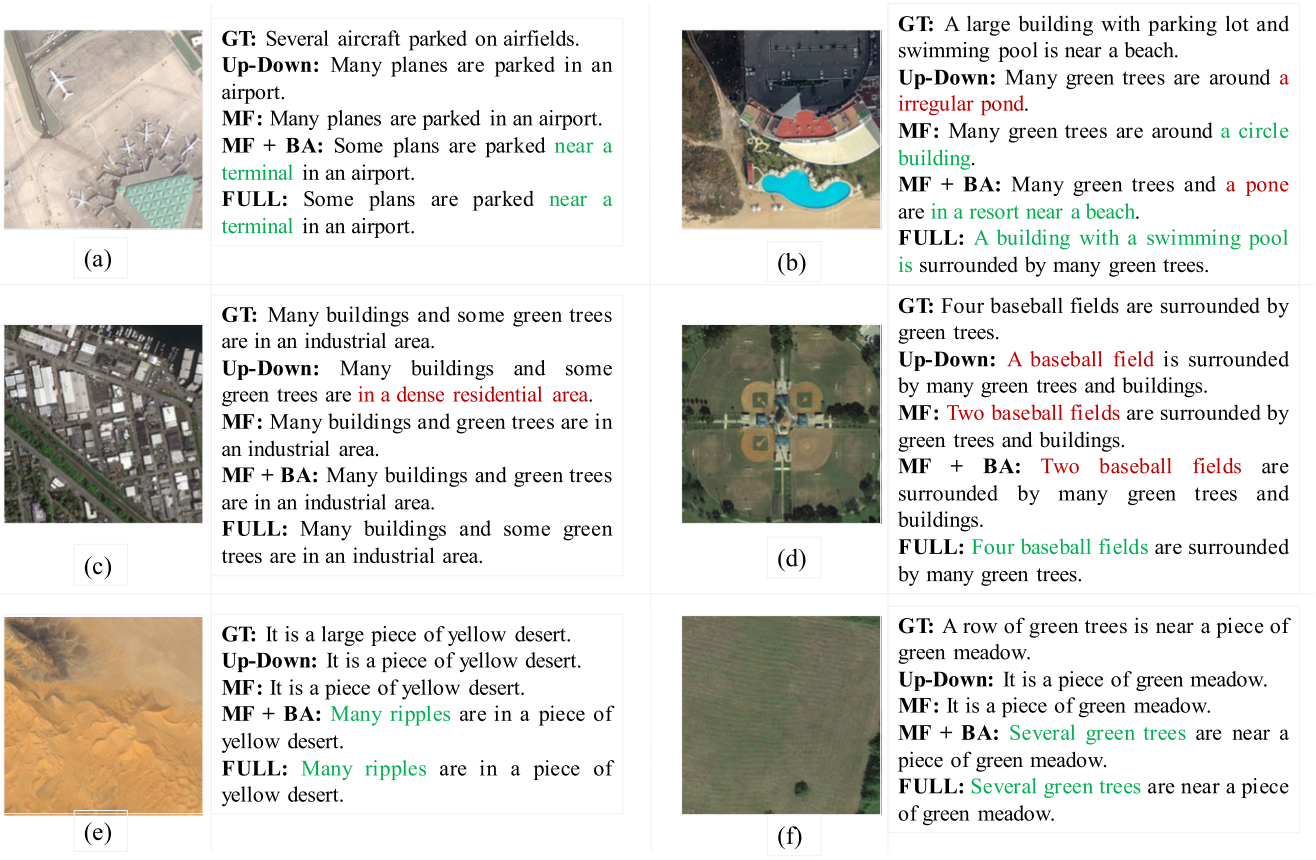


Fig. 7. Some typical examples on the RSICD test set. The GT sentences are human-annotated sentences, and the other sentences are generated by the ablation models. The wrong words generated by all models are signed with red font; the green font words are generated by the models with our boosted attention.

TABLE VII
ABLATION PERFORMANCE OF OUR DESIGNED MODEL ON RSICD DATASET [9]

Methods	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
Up-Down [26]	0.7708	0.6542	0.5614	0.4848	0.3952	0.6580	2.6237	0.4658	0.9183
EF	0.7752	0.6586	0.5647	0.4867	0.3622	0.6650	2.6438	0.4678	0.9251
EF + RA	0.7737	0.6609	0.5716	0.4988	0.3640	0.6629	2.7202	0.4709	0.9434
FULL	0.7729	0.6651	0.5782	0.5062	0.3626	0.6691	2.7549	0.4719	0.9529

TABLE VIII
PERFORMANCE OF OUR MODEL UNDER DIFFERENT DILATION RATES ON RSICD DATASET [9]

Dilation Rate	Bleu1	Bleu2	Bleu3	Bleu4	Meteor	Rouge	Cider	Spice	S_m
2,4,6,8	0.7696	0.6611	0.574	0.5011	0.3604	0.6634	2.6945	0.4698	0.9379
2,4,8,16	**0.7729**	**0.6651**	**0.5782**	**0.5062**	**0.3626**	**0.6691**	**2.7549**	**0.4719**	**0.9529**
2,8,16,32	0.7619	0.6491	0.559	0.4858	0.3635	0.666	2.6557	0.467	0.9276

field, while the original CNN features concern about whole spatial scene information. They all should be considered for better remote sensing image representation. Secondly, due to introducing the "visual sentinel" concept and recurrent attention network, the EF + RA model further improves the performance of the EF model. This phenomenon indicates that the usage of multimodal features of linguistic knowledge and visual features is important and necessary for visual content understanding. Finally, our full model that adds a semantic gate to enhance the semantic vector output from the Language LSTM, is an improved version of the EF + RA model. As can be seen in Table VII, the semantic gate achieves great success, because the gate integrates different semantic branches effectively, which is very helpful for enhancing word state representations more comprehensively and precisely.

2) Qualitative Comparison: As discussed above, the ablation models play a positive role in the remote sensing image understanding. To further demonstrate the discriminability, we show some qualitative results on the test set of the RSICD dataset, as shown in Fig. 7. The EF model without the recurrent

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attention may lost some potential object information in the process of generating description. For example, the EF model can describe the main object in the scene (such as "planes" and "airport"), but the "terminal" is not been described, as shown in Fig. 7(a). Fig. 7(e) is similar with Fig. 7(a) losing the "ripples" in the description, two main reasons of which include that: 1) the image reference information provides less information of some targets for predicting sentences and 2) the EF model does not have the guidance of visual information which makes the model focus on relevant visual areas. In other words, the EF model only focuses on the main region (the "baseball fields" and "meadow" region) and describes it. Though the EF + RA model makes full use of images to provide more necessary visual features, there are still some weaknesses. We find from Fig. 7(d) that the generated sentences from Up-Down, EF, and EF + RA are weak in quantitative judgment. Although the sentence reference provides some prior knowledge, it is difficult for the Up-Down, EF, and EF + RA models to draw such an accurate quantitative information. The FULL can tackle this problem and generate the accurate caption—"four baseball fields." The semantic gate can correct some mistakes in the EF + RA. The FULL model identifies precisely some object as "a swimming pool" but not "a pond," as shown in Fig. 7(b), which indicates that our proposed RASG model not only pays attention to the comprehensive object information but also digs important semantic information to generated the description. Therefore, the semantic of the remote sensing images can be represented more comprehensively and precisely by our proposed RASG model.

G. Parameters Analysis

Our model contains multiple important parameters, which may influence the effectiveness of remote sensing image captioning. In this section, we analyze different dilation rates used in ESP module. This parameter plays a crucial role in dilated convolutions. The larger the dilation rate is set, the larger the receptive field is extracted. In this experiment, the dilation rates are set to (2, 4, 6, 8), (2, 4, 8, 16), and (2, 8, 16, 32) respectively for comparison, while the original CNN features are extracted by VGG16. As depicted in Table VIII, the performance of our proposed approach on RSICD dataset obviously changes along with the variation of dilation rates.

In Table VIII, we can see that the best results are obtained when dilation rates are set to (2, 4, 8, 16). A smaller set of dilation rates allows the ESP module to focus on fewer context information and relational information with smaller receptive fields, which may cause weaker performances. For the greater set of dilation rates, the performance decreases drastically with some fluctuations. The reason is that the ESP module will abandon a lot of detailed information of remote sensing images and have a negative impact on the model, which leads to a decline in performance. As a whole, the blind increase of the dilation rates does not benefit in improving the performance. Based on this observation, the dilation rates (2, 4, 8, 16) are available for our model.

TABLE IX
COMPARISON BETWEEN OUR METHOD AND THE UP-DOWN MODEL ON THE NUMBER OF PARAMETERS, FLPOS, TRAINING TIME, AND INFERENCE SPEED (IMAGES PER SECOND). ALL RESULTS ARE REPORTED BASED ON RSICD DATASET [9]

Methods	#parameters	FLOPs	training time	Inference speed
Up-Down [26]	45.18	15.89G	266min	16.73
Ours	53.44	15.95G	228min	16.50

H. Speed Performance

Similar to [33], we also compare the other four aspects with the Up-Down model on RSICD dataset. The four aspects contain the number of model parameters, the number of floating-point operations (FLOPs), training time, and the inference speed (images per second). The comparison results between our method and the Up-Down model are shown in Table IX. Particularly, the training time and inference speed are tested on single Nvidia GeForce GTX 1080Ti GPU, and the CNN framework is VGG16 which is same in two models. Comparing with the Up-Down method, the number of model parameters and the number of FLOPs in our model are more than that of the Up-Down model. This is because we modify the Up-Down model with three innovative components including feature extractor, recurrent attention, and semantic gate. However, the less time is used in our training process due to the superiority of the designed model. In addition, because of the increase in the number of parameters, the inference time shows slightly more than the Up-Down model.

V. CONCLUSION

In this article, we present a novel approach to generate the natural-language description for the remote sensing image. At first, the encoded competitive features are obtained by feature extractor which take advantage of the original CNN features and multiscale features. Moreover, a recurrent attention algorithm is designed to detect the visually and semantically salient vector based on encoded competitive features and nonvisual features. Thus, the ability of the Language LSTM is benefit from the high-level contextual vector. Furthermore, we also explore how to make full use of the different semantic representations with a semantic gate. It adopted the "visual sentinel" vector of the Language LSTM, the hidden states of two LSTM layers and word embedding at t time. Finally, experimental results on three benchmark datasets demonstrate superior performance over several other state-of-the-art methods.

In the future, we plan to upgrade our remote sensing image captioning model on three main directions: 1) exploring more effective encoding techniques for different characteristics of the remote sensing images; 2) measuring implicit relationships among different objects in each image; and 3) evaluating the proposed model with CIDEr optimization.

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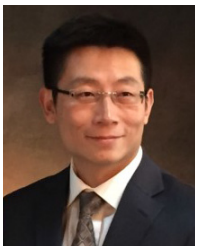
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